

KEMI TORNIO, Finland

WMO: 028640

Lat: 65.784N

Lon: 24.579E

Elev: 61

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-12.8	-23.7	1.5	-17.9	-17.7	2.1	-12.6	25.4	28.1	22.7	28.3	3.5	10	0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	76.7	63.7	73.1	62.3	69.9	60.8	66.5	73.5	64.2	70.6	62.2	67.8	8.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	89.6	69.9	61.6	82.0	67.0	59.5	76.1	64.7	31.1	72.9	29.4	70.8	28.0	68.1	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.1	18.7	16.6	-23.0	80.6	5.9	4.4	-27.2	83.8	-30.7	86.4	-34.0	88.8	-38.3	92.1		
			-23.2	68.4	5.8	3.4	-27.4	70.8	-30.8	72.8	-34.1	74.7	-38.3	77.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	13.7	14.2	21.7	33.1	44.1	53.5	60.4	56.5	48.3	36.4	27.0	21.0
	DBStd	18.27	13.47	13.46	9.73	5.98	7.48	5.84	5.49	5.87	6.24	7.57	9.94	11.93
	HDD50	5888	1126	1002	878	509	215	28	2	14	105	421	690	898
	HDD65	10631	1591	1422	1343	958	649	348	161	269	502	885	1140	1363
	CDD50	761	0	0	0	0	32	134	325	216	53	1	0	0
	CDD65	30	0	0	0	0	1	4	19	6	0	0	0	0
	CDH74	242	0	0	0	0	20	35	162	25	0	0	0	0
	CDH80	35	0	0	0	0	1	7	26	1	0	0	0	0
	WSAvg	7.5	7.2	7.4	7.6	7.3	7.9	7.8	7.0	7.2	7.6	7.6	7.7	7.5
	PrecAvg	23.7	2.0	1.6	1.4	1.2	1.6	1.8	2.8	2.4	2.1	2.2	2.4	2.0
Precipitation	PrecMax	33.0	3.6	3.6	2.5	3.2	2.7	3.1	5.2	5.9	5.0	4.3	4.4	4.7
	PrecMin	18.0	0.5	0.2	0.4	0.3	0.4	0.4	0.4	0.5	0.6	0.8	0.7	0.6
	PrecStd	4.6	0.9	0.9	0.6	0.8	0.6	0.9	1.3	1.4	1.2	0.9	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.3	40.5	42.9	55.3	76.8	79.1	83.9	77.3	65.3	53.3	45.6	39.8
		MCWB	34.9	36.0	35.9	44.0	58.2	64.3	66.9	64.8	57.6	49.9	44.0	38.4
	2%	DB	34.1	35.9	39.2	49.6	68.1	72.0	78.2	73.1	61.9	50.9	43.9	37.7
		MCWB	33.5	34.1	34.8	42.0	54.4	59.6	64.6	62.9	55.9	48.7	42.9	36.6
	5%	DB	32.9	33.7	35.9	45.2	61.7	68.1	74.7	69.8	59.6	49.1	41.1	36.0
		MCWB	32.5	33.0	33.3	39.1	51.0	56.8	63.6	61.7	55.4	47.1	40.2	35.0
	10%	DB	31.7	31.9	34.0	42.0	57.1	64.6	71.6	66.7	57.8	46.8	39.2	34.7
		MCWB	31.3	31.4	32.2	36.9	48.7	55.3	62.5	60.5	54.4	45.1	38.3	33.9
	0.4%	WB	35.0	36.9	37.4	46.5	59.7	66.0	69.9	68.7	59.6	51.3	44.8	38.9
		MCDB	35.7	39.4	40.9	51.1	72.3	75.3	77.0	74.0	63.0	52.3	45.1	39.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.7	34.4	35.2	42.5	55.9	62.2	67.4	65.3	57.7	49.3	42.7	36.8
		MCDB	34.1	35.5	37.6	48.5	66.4	70.0	74.4	70.1	60.3	50.6	43.4	37.4
	5%	WB	32.4	32.9	33.9	39.6	52.3	59.3	65.5	63.1	56.5	47.3	40.2	35.3
		MCDB	32.7	33.5	35.4	44.3	59.9	64.9	72.9	67.2	59.0	48.6	40.9	35.9
	10%	WB	31.1	31.5	32.5	37.5	49.4	56.9	63.4	61.3	55.1	45.3	38.3	33.9
		MCDB	31.4	32.0	33.8	40.9	55.7	62.3	70.0	65.7	57.5	46.6	39.0	34.5
Mean Daily Temperature Range	5% DB	MDBR	13.2	14.1	15.9	13.7	15.9	15.2	16.2	15.0	13.1	10.4	9.8	11.4
		MCDBR	10.2	10.3	13.4	18.4	24.0	21.7	22.0	18.7	14.9	11.0	8.5	8.7
		MCWBR	10.1	9.3	10.5	12.2	13.1	11.1	11.5	10.8	10.4	9.4	8.4	8.3
	5% WB	MCDBR	10.5	9.9	11.3	16.8	21.6	15.7	18.2	14.7	11.8	9.9	8.0	8.2
		MCWBR	10.4	9.1	9.2	11.7	12.8	9.8	10.7	9.7	9.0	9.0	8.2	8.0
Clear-Sky Solar Irradiance	taub		0.206	0.273	0.302	0.339	0.345	0.349	0.366	0.350	0.327	0.302	0.344	0.356
	taud		1.944	2.095	2.151	2.214	2.422	2.478	2.451	2.483	2.528	2.347	2.573	2.612
	Ebn at Noon		78	184	239	260	272	274	265	258	239	179	75	24
	Edh at Noon		6	18	28	34	31	30	30	26	20	15	6	3
All-Sky Solar Radiation	RadAvg		26	178	644	1192	1557	1743	1623	1184	631	226	44	5
	RadStd		2	35	65	95	141	132	143	108	67	39	7	0

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page